

ELECTRON CONFIGURATION WORKSHEET

1. What is the subshell designation (e.g. 2p, 3d...) for the following cases?

a) $n = 2, l = 0$ _____ 2s _____

b) $n = 4, l = 3$ _____ 4f _____

c) $n = 5, l = 1$ _____ 5p _____

d) $n = 3, l = 0$ _____ 3s _____

e) $n = 6, l = 1$ _____ 6p _____

f) $n = 5, l = 2$ _____ 5d _____

2. The quantum numbers listed below are for 4 different electrons in the same atom. Arrange them in order of increasing energy.

a) $n = 4, l = 0, m_l = 0, m_s = \frac{1}{2}$

_____ a _____ least energy

b) $n = 3, l = 2, m_l = 1, m_s = \frac{1}{2}$

_____ d _____

c) $n = 3, l = 2, m_l = -1, m_s = \frac{1}{2}$

_____ b _____

d) $n = 3, l = 1, m_l = 1, m_s = -\frac{1}{2}$

_____ c _____ highest energy

Do any have the same energy? _____ Yes _____ which ones? _____ b/c _____

3. Write the complete electron configuration (simplified) for the following, using only the periodic table as a guide (do not use noble gas core configuration):

a) Cd _____ 1s² 2s² 2p⁶ 3s² 3p⁶ 4s² 3d¹⁰ 4p⁶ 5s² 4d¹⁰ _____

b) As _____ 1s² 2s² 2p⁶ 3s² 3p⁶ 4s² 3d¹⁰ 4p³ _____

c) Sr _____ 1s² 2s² 2p⁶ 3s² 3p⁶ 4s² 3d¹⁰ 4p⁶ 5s² _____

d) Sb _____ 1s² 2s² 2p⁶ 3s² 3p⁶ 4s² 3d¹⁰ 4p⁶ 5s² 4d¹⁰ 5p³ _____

e) S _____ 1s² 2s² 2p⁶ 3s² 3p⁴ _____

4. Write the outer shell electron configurations for the following, using the periodic table as a guide (this means use the noble gas core configurations)

a) K _____ [Ar] 4s¹ _____

b) Al _____ [Ne] 3s² 3p¹ _____

- c) Fe _____ [Ar] 4s² 3d⁶ _____
- d) Bi _____ [Xe] 6s² 4f¹⁴ 5d¹⁰ 6p³ _____
- e) F _____ [He] 2s² 2p⁵ _____

5. Give outer-shell configuration for the following ions (Noble gas core):

- a) Te²⁻ _____ 52 + 2 = 54 e⁻: [Xe] _____
- b) Rb⁺ _____ 37 - 1 = 36 e⁻: [Kr] _____
- c) Br⁻ _____ 35 + 1 = 36 e⁻: [Kr] _____
- d) Mn²⁺ _____ [Ar] 3d⁵ _____
- e) Cr³⁺ _____ [Ar] 3d³ _____

6. Which has the LARGER ionization energy?

- a) B or **Cl** b) **N** or P c) **Hf** or Cs
- d) Ga or **Ge** e) K or **K⁺** f) Cs⁺ or **Ba⁺**

7. Which has the most EXOTHERMIC electron affinity?

- a) **Cl** or Ar b) Se or **Br** c) **Si** or P
- d) Fr or **F** e) **Se** or Se⁻ f) **Pb** or Po

8. Which of the following are DIAMAGNETIC?

- a) Sn _____ b) Hg _____ diamagnetic _____
- c) Sr _____ diamagnetic _____ d) Sb _____
- e) Ir _____ f) Br⁻ _____ diamagnetic _____

9. What is the Pauli exclusion principle?

Pauli exclusion principle basically says you can't have two electrons in the same atom with the exact same four quantum numbers. So in one orbital, you can only fit two electrons max and they have to have opposite spins.